

Coordination-Aware Ergodic Coverage for Multi-UAV Search and Rescue

Inayyah Don Nazwim

Electrical Engineering and Computer Sciences

University of California, Berkeley

Berkeley, USA

inayyah@berkeley.edu

Abstract—Multi-UAV search and rescue (SAR) systems require efficient spatial coverage of partially observable environments. Mendoza et al. [2] propose a decentralized ergodic coverage framework in which each UAV maintains a GP-based belief over an unknown importance map and computes an ergodic Markov-chain policy to distribute sensing effort proportionally to estimated importance. While this framework achieves individual ergodic coverage, the authors identify a key limitation: UAVs computing policies from similar beliefs produce similar policies, leading to spatial redundancy. We introduce a coordination-aware modification that adjusts each UAV’s target distribution by subtracting a weighted version of its neighbors’ empirical visitation before policy computation, requiring no additional communication. We propose two formulations—uniform subtraction (V1) and an importance-weighted refinement (V2)—and evaluate both against the baseline on regret, spatial overlap, and ROI discovery time, with additional studies on belief update frequency and team size. V2 with $\alpha = 0.2$ reduces spatial overlap by 73%, improves last-ROI discovery time by 32%, and maintains sublinear regret within 3% of the baseline. Code: https://github.com/inayyah-d/290_final_project_code.

I. INTRODUCTION

Unmanned aerial vehicles are increasingly deployed in disaster response, where they provide aerial coverage, situational awareness, and victim localization in hazardous environments that are difficult for human responders to access [1]. Despite this promise, most current deployments remain heavily human-supervised, with operators managing individual drones and relying on pre-planned trajectories whose data becomes available only hours after collection. The goal is to transition toward systems where one operator oversees many autonomous drones simultaneously.

This transition introduces a fundamental coordination challenge. When multiple UAVs operate without explicit coordination, they independently optimize for similar objectives, leading to redundant coverage of high-importance regions while other areas of the map go unobserved. In SAR applications, this inefficiency directly translates to delayed victim discovery and increased mission time.

Recent work by Mendoza et al. [2] addresses the individual coverage problem through an ergodic framework: each UAV computes a Markov-chain policy whose stationary distribution matches an estimated importance map, naturally balancing exploration of unknown regions with repeated monitoring of high-priority areas. The framework is decentralized and

robust to communication constraints. However, the authors explicitly identify that the current policy does not account for coordination across agents—UAVs with similar beliefs compute similar policies and cluster at the same regions.

Contributions. This paper addresses this gap with the following contributions:

- We introduce a coordination-aware modification to the ergodic coverage policy that personalizes each UAV’s target distribution by subtracting a weighted neighbor empirical coverage signal before policy computation (Section III).
- We propose two formulations: V1 (uniform subtraction) and V2 (importance-weighted subtraction), where V2 protects high-importance regions from being down-weighted.
- We evaluate both formulations against the baseline on regret, ROI discovery time, spatial overlap, belief update frequency sensitivity, and scalability across team sizes (Section IV).

II. RELATED WORK

A. RESCUE Framework

Mendoza et al. [1] model multi-UAV SAR as a Partially Observable Markov Decision Process (POMDP) with two heterogeneous UAV roles: a High-Level Rescuer (HLR) that provides aerial surveillance and a Low-Level Rescuer (LLR) that physically intercepts evacuees. Policies are learned via Proximal Policy Optimization (PPO) with recurrent networks. The framework assumes perfect inter-UAV communication and handles only a single evacuee; crucially, it provides no explicit coordination strategy for multiple HLRs, leaving open the question of how to prevent redundant coverage among aerial agents. Our work directly addresses this HLR coordination gap.

B. Decentralized Ergodic Coverage

Mendoza et al. [2] propose the framework we extend. Each UAV fits a Gaussian Process to its observation dataset, constructs a UCB-based belief target distribution, and computes a transition matrix via REMC [3] that minimizes the second-largest eigenvalue $\lambda_2(P)$, maximizing mixing speed. The framework achieves robustness to limited communication but explicitly leaves inter-agent policy coordination as future

work. Our modification targets this gap at the level of target distribution construction.

C. Multi-Agent Ergodic Coverage

Several works address multi-agent ergodic coordination with different approaches. Abraham and Murphey [8] derive a decentralized ergodic control policy for agents with nonlinear dynamics that uses consensus to coordinate coverage; they show that any inter-agent communication improves convergence and establish Nash equilibrium properties for the resulting policies. However, their formulation requires a shared, known target distribution rather than inferring one online. Gkouletsos et al. [9] extend ergodic multi-agent planning to settings with limited communication by optimizing a joint cost over ergodicity, control energy, and inter-agent proximity, showing improved completion time through agent collaboration; their approach is centralized in cost computation and does not handle unknown environments. Our method differs from both: we maintain a fully decentralized architecture, operate over an unknown importance map inferred via GP, and introduce coordination at the target distribution level rather than through trajectory optimization or consensus dynamics.

Prattisoli et al. [10] address a closely related problem: distributed coverage control over time-varying spatial fields using GP models. Their approach dynamically balances exploration and exploitation and manages data volume for computational tractability. Relative to their work, our method focuses on the discrete Markov-chain setting of [2] and introduces an explicit coordination term to prevent spatial redundancy, which they do not address.

D. Coverage Control

Classical coverage control [4] positions robots to minimize a cost function over a domain. Extensions handle heterogeneous sensing [5] and time-varying densities [6]. These formulations typically assume known density functions and optimize static configurations, whereas our setting involves unknown, dynamic importance maps that must be inferred online.

E. Multi-Robot Coordination

Karam et al. [7] distinguish cooperation (shared goal), coordination (coupled decisions), and collaboration (joint capability) in multi-robot systems. Under this taxonomy, the Mendoza ergodic framework achieves coordination via belief sharing but not via coupled policy computation, where each UAV independently runs REMC without considering teammates' planned movements. Our modification introduces policy-level coordination while preserving decentralized execution.

III. PROBLEM FORMULATION AND METHODS

A. Environment

We represent the environment as an undirected graph $\mathcal{G} = (\mathcal{R}, \mathcal{E})$ with $|\mathcal{R}| = N$ regions, where $(r_i, r_j) \in \mathcal{E}$ if regions r_i and r_j are adjacent. Each region $r_i \in \mathcal{R}$ is assigned a non-negative, time-varying weight $w_k(r_i) \geq 0$ reflecting its

information content. The weights form the *information map* $\phi_k^* \in \mathbb{R}_{\geq 0}^N$. The *true target distribution* is the normalized map:

$$\rho_k^* = \frac{\phi_k^*}{\mathbf{1}^\top \phi_k^*} \quad (1)$$

The map evolves according to exogenous change times $\mathcal{K} \subset \{1, \dots, K\}$:

$$\phi_k^* = \begin{cases} \phi_{k-1}^*, & k \notin \mathcal{K} \\ \mathcal{U}_k(\phi_{k-1}^*), & k \in \mathcal{K} \end{cases} \quad (2)$$

where $\mathcal{U}_k : \mathbb{R}_{\geq 0}^N \rightarrow \mathbb{R}_{\geq 0}^N$ is an exogenous transformation (e.g., fire spread or population relocation). Both the initial map ϕ_0^* and change times $\mathcal{K}$ are unknown to the UAVs a priori.

B. UAV Agents

A team of M UAVs $\mathcal{M} = \{1, \dots, M\}$ operates autonomously. At timestep k , UAV m occupies region $x_k^m \in \mathcal{R}$ and collects noisy observations within sensing range R_{sense} :

$$Y_k^m(x_k^m) \triangleq \{(x, \phi_{k,r}^* |_{r=x} + \varepsilon_k^m) \mid x \in \text{Ball}_{\mathcal{R}}(x_k^m, R_{\text{sense}})\} \quad (3)$$

where ε_k^m is sensor noise. The *empirical visitation distribution* of UAV m is:

$$\hat{\rho}_k^m = \frac{1}{k+1} \sum_{\tau=0}^k \mathbb{I}_{\mathcal{R}}(x_\tau^m) \quad (4)$$

where $\mathbb{I}_{\mathcal{R}}(x_\tau^m) \in \{0, 1\}^N$ is a one-hot indicator. The *team empirical distribution* is $\hat{\rho}_k \triangleq \frac{1}{M} \sum_{m=1}^M \hat{\rho}_k^m$.

C. Objective

The team minimizes time-averaged regret:

$$\text{Regret}(K) = \frac{1}{K} \sum_{k=1}^K \mathbb{E} [\|\hat{\rho}_k - \rho_k^*\|_1] \quad (5)$$

We desire sublinear regret: $\text{Regret}(K) \leq \mathcal{O}(K^\alpha + \mathcal{V})$ where $\alpha < 0$ and $\mathcal{V} = \lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=1}^K \|\rho_k^* - \rho_{k-1}^*\|$ captures the long-run average change rate of the environment.

D. Baseline: Decentralized Ergodic Coverage

Algorithm 1 summarizes the Mendoza et al. framework. Each UAV independently maintains a dataset $\mathcal{D}_k^m$ aggregating its own observations and those shared by communication neighbors $\mathcal{N}_k^m = \{l \in \mathcal{M} \setminus \{m\} \mid \|x_k^m - x_k^l\|_2 \leq R_{\text{comm}}\}$. Every τ_{GP} steps, it fits a GP to $\mathcal{D}_k^m$ and applies UCB to construct a belief target:

$$\bar{\phi}_k^m(r) \triangleq \mu_k^m(r) + \beta \sigma_k^m(r) \quad (6)$$

where μ_k^m and σ_k^m are the GP posterior mean and standard deviation, and $\beta > 0$ balances exploration and exploitation. The belief target distribution $\bar{\rho}_k^m$ is the normalization of $\bar{\phi}_k^m$. Every τ_P steps, each UAV calls REMC [3] on its belief target to obtain a transition matrix P_k^m whose stationary distribution is $\bar{\rho}_k^m$ and which minimizes $\lambda_2(P)$ for fast mixing. The UAV then samples its next region from P_k^m .

Algorithm 1 Mendoza et al. Algorithm 1: Decentralized Ergodic Exploration

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1: Input:  $T_{\text{final}}, \tau_{\text{GP}}, \tau_P, R_{\text{comm}}, R_{\text{sense}}, x_1^m$ 
2: Init:  $\bar{\rho}_0^m \leftarrow \frac{1}{n} \mathbf{1}, \mathcal{D}_0^m \leftarrow \emptyset, P_0^m \leftarrow \text{REMC}(\bar{\rho}_0)$ 
3: for  $k = 1, \dots, T_{\text{final}} - 1$  do
4:    $Y_k^m \leftarrow \phi_k^*(x_k^m) + \varepsilon_k^m$  ▷ Observe
5:    $\mathcal{D}_k^m \leftarrow \mathcal{D}_{k-1}^m \cup Y_k^m$ 
6:    $\mathcal{N}_k^m \leftarrow \{l \in \mathcal{M} \setminus \{m\} \mid \|x_k^m - x_k^l\|_2 \leq R_{\text{comm}}\}$ 
7:   for each  $l \in \mathcal{N}_k^m$  do
8:      $\mathcal{D}_k^m \leftarrow \mathcal{D}_k^m \cup Y_k^l$  ▷ Share
9:   end for
10:  if  $\text{mod}(k, \tau_{\text{GP}}) = 0$  then
11:     $\bar{\phi}_k^m, \bar{\rho}_k^m \leftarrow \mathcal{GP}_{\text{UCB}}(\mathcal{D}_k^m)$  ▷ Line 11
12:  else
13:     $\bar{\rho}_k^m \leftarrow \bar{\rho}_{k-1}^m$ 
14:  end if
15:  if  $\text{mod}(k, \tau_P) = 0$  then
16:     $P_k^m \leftarrow \text{REMC}(\bar{\rho}_k^m)$ 
17:  else
18:     $P_k^m \leftarrow P_{k-1}^m$ 
19:  end if
20:   $x_{k+1}^m \leftarrow x \sim P_k^m(\cdot \mid x_k^m)$  ▷ Move
21: end for

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E. The Coordination Gap

Because all UAVs share observations within R_{comm} , their datasets $\mathcal{D}_k^m$ are similar $\Rightarrow$ similar GP posteriors $\mu_k^m, \sigma_k^m \Rightarrow$ similar UCB outputs $\bar{\phi}_k^m \Rightarrow$ similar belief targets $\bar{\rho}_k^m \Rightarrow$ similar REMC policies P_k^m . The result is that all M UAVs tend to cluster at the same high-weight regions, increasing spatial overlap and leaving low-weight regions persistently under-covered. There is no mechanism in Algorithm 1 that says: *my teammate is covering region r , so I should go elsewhere*.

F. Coordination-Aware Modification

We modify Line 11 of Algorithm 1 to produce a personalized target distribution for each UAV that accounts for where teammates are already spending time. Let $\hat{\rho}_k^j$ denote the empirical visitation distribution of neighbor $j \in \mathcal{N}_k^m$, already available from the communication step (lines 6–9).

V1 — Uniform Subtraction. Our initial formulation subtracts the average neighbor empirical coverage uniformly:

$$\bar{\phi}_{k,\text{coord}}^m(r) = \bar{\phi}_k^m(r) - \alpha \cdot \frac{1}{|\mathcal{N}_k^m|} \sum_{j \in \mathcal{N}_k^m} \hat{\rho}_k^j(r) \cdot \|\bar{\phi}_k^m\|_1 \quad (7)$$

V2 — Importance-Weighted Subtraction. V1 subtracts uniformly regardless of region importance, which can push UAVs away from high-importance ROIs. V2 addresses this by weighting the subtraction by $(1 - \bar{\rho}_k^m(r))$, protecting high-importance regions:

$$\bar{\phi}_{k,\text{coord}}^m(r) = \bar{\phi}_k^m(r) - \alpha (1 - \bar{\rho}_k^m(r)) \cdot \frac{1}{|\mathcal{N}_k^m|} \sum_{j \in \mathcal{N}_k^m} \hat{\rho}_k^j(r) \cdot \|\bar{\phi}_k^m\|_1 \quad (8)$$

For both formulations, the modified target distribution is:

$$\bar{\rho}_k^m \leftarrow \text{normalize}(\text{clip}(\bar{\phi}_{k,\text{coord}}^m, 10^{-8})) \quad (9)$$

Properties.

- *Decentralized:* uses only $\hat{\rho}_k^j$ from $\mathcal{N}_k^m$, already shared in lines 6–9. No new communication required.
- *Validity:* clipping ensures non-negativity; normalization ensures sum to 1. $\bar{\rho}_k^m$ is always a valid REMC input, preserving the ergodic structure.
- *Reduces to baseline:* when $\alpha = 0$, both V1 and V2 recover Algorithm 1 exactly.

The coordination parameter $\alpha \in [0, 1]$ controls how aggressively UAVs avoid regions covered by neighbors. We select α empirically via ablation (Section IV).

REMC Implementation. We implement REMC using the Metropolis-Hastings construction [3], a closed-form method that produces a valid ergodic Markov chain with the desired stationary distribution without requiring convex optimization at each step. The ergodic property is preserved exactly; the only difference from the full REMC optimum is potentially slower mixing.

IV. EXPERIMENTS

A. Setup

We simulate on an 8×8 grid ($N = 64$ regions) with $M = 3$ UAVs, four Gaussian ROI centers at positions (2, 2), (2, 5), (5, 2), (5, 5), and a 2×2 no-fly zone at the center. All experiments use $T = 600$ timesteps, $\tau_{\text{GP}} = \tau_P = 20$, $R_{\text{comm}} = 2.5$, $R_{\text{sense}} = 1.5$, and $\beta = 1.0$. We compare three conditions: **Baseline** ($\alpha = 0$, Algorithm 1 unmodified), **V1** ($\alpha = 0.2$, Eq. 7), and **V2** ($\alpha = 0.2$, Eq. 8). All results are averaged over 5 independent random seeds; shaded regions show ± 1 standard deviation.

The $\alpha = 0.2$ value was selected based on a preliminary sweep (Section IV-C) showing it achieves the best tradeoff between regret and ROI discovery time.

Metrics. We evaluate efficiency along two complementary dimensions: *spatial redundancy* and *coverage completeness*. Spatial redundancy is quantified by spatial overlap, which is the fraction of timesteps where at least two UAVs co-occupy a region. High overlap indicates that agents are duplicating sensing effort, which directly wastes sensing capacity. Coverage completeness is measured by ROI discovery time, specifically the time to discover the *last* ROI, since in SAR the mission is incomplete until every high-importance region has been visited. We additionally track time-averaged regret (Eq. 5) and belief error $\mathcal{L}_{\text{belief},k} = \|\bar{\rho}_k - \rho_k^*\|_1$ to assess whether coordination imposes an unacceptable cost on individual coverage quality.

B. Main Results

Regret. Figure 1 shows that all three conditions achieve sublinear regret, consistent with the desired $\mathcal{O}(K^\alpha)$ bound. The coordinated variants incur a small regret penalty ($\sim 3\%$), which reflects a deliberate tradeoff: the coordination term

reduces sensing redundancy by pushing UAVs toward less-visited regions, some of which carry lower importance. V2 achieves lower regret than V1, confirming that importance-weighting better preserves ergodic coverage quality while still diversifying coverage.

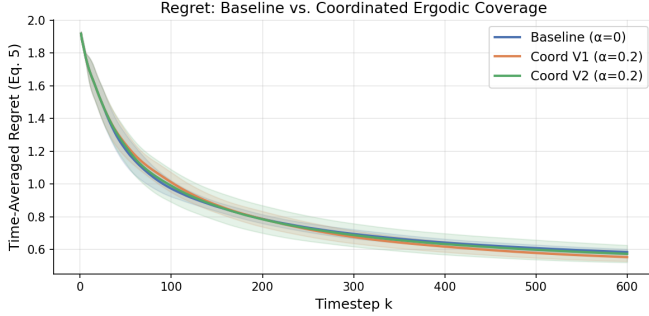


Fig. 1. Time-averaged regret (Eq. 5) for baseline, V1, and V2 with $\alpha = 0.2$. All conditions achieve sublinear regret. The $\sim 3\%$ penalty reflects the tradeoff between individual coverage quality and spatial diversification.

Spatial Overlap and Belief Error. Figure 2 presents the central efficiency result. Spatial overlap, our direct measure of sensing redundancy, is reduced by 73% under V2 relative to the baseline (from 0.026 to 0.007). Concretely, the baseline has two UAVs co-located in roughly 1 in 38 timesteps, wasting that agent’s sensing capacity; V2 reduces this to 1 in 143 timesteps. This freed capacity is redirected to less-visited regions, enabling more even coverage and directly enabling faster full-area discovery. Belief error is slightly higher under coordination, as personalized targets deliberately diverge from the shared GP estimate.

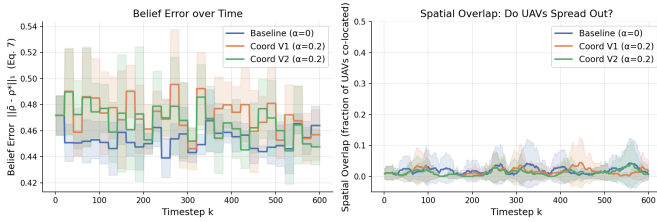


Fig. 2. Belief error (left) and spatial overlap (right) over time. V2 reduces spatial overlap by 73%, substantially reducing wasted sensing capacity.

ROI Discovery. Figure 3 shows how reduced spatial redundancy translates to coverage completeness. Under the baseline, sensing capacity wasted on co-located agents is capacity unavailable to discover the last ROI: the baseline’s last-ROI discovery time is 60.8 timesteps, while V2 reduces this to 41.0—a 32% improvement. Mean discovery time across all four ROIs is comparable across conditions (30.0, 29.5, 30.9 for baseline, V1, V2 respectively), confirming that coordination does not impede discovery of any individual ROI; its effect is specifically to eliminate the long tail caused by spatial clustering.

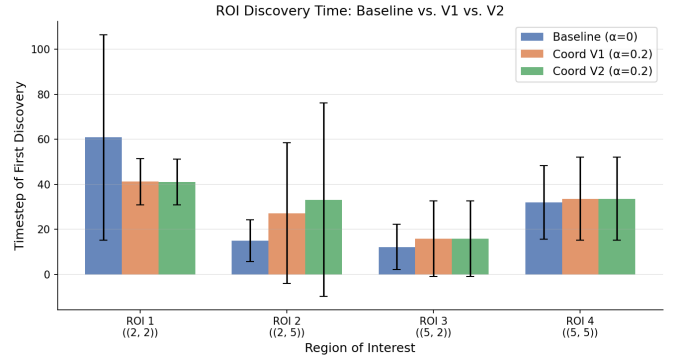


Fig. 3. ROI discovery time for each of the four ROI centers. Coordination spreads discovery more evenly, reducing last-ROI discovery time by 32% with comparable mean discovery time across all ROIs.

Coverage Heatmaps. Figure 4 visualizes final empirical coverage. The baseline over-concentrates on certain ROIs. V1 introduces coverage artifacts—regions visited more than their importance warrants, visible as warm patches not corresponding to the true target ρ^* . V2 reduces these artifacts through importance-weighting, producing empirical coverage that more closely tracks ρ^* while still achieving spatial spread.

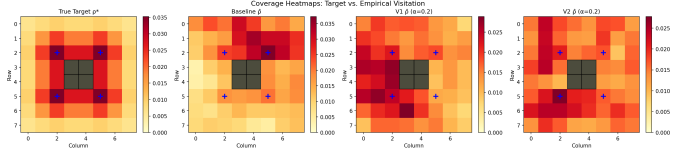


Fig. 4. Coverage heatmaps. From left: true target ρ^* , baseline $\hat{\rho}$, V1 $\hat{\rho}$, V2 $\hat{\rho}$. Blue crosses mark ROI centers. V2 reduces the coverage artifact present in V1.

Table I summarizes key numerical results across conditions.

TABLE I
SUMMARY OF KEY METRICS ($M = 3$, $\alpha = 0.2$, $\tau = 20$, AVERAGED OVER 5 SEEDS)

Metric	Baseline	V1	V2
Final Regret	0.596	0.610	0.604
Mean ROI Discovery (steps)	30.0	29.5	30.9
Last ROI Discovery (steps)	60.8	41.2	41.0
Mean Spatial Overlap	0.026	0.009	0.007

C. Alpha Selection

We first conducted a preliminary sweep of $\alpha \in \{0.0, 0.2, 0.4, 0.6, 0.8\}$ comparing the baseline against V1, measuring final regret and spatial overlap. V1 showed reduced regret at $\alpha \in \{0.2, 0.4\}$ with degradation at $\alpha = 0.8$, suggesting this as the operating range. We then evaluated both α values under the belief update frequency experiment (Section IV-D) using V2, and found $\alpha = 0.2$ achieves lower regret and faster last ROI discovery across all update periods. All primary comparisons therefore use $\alpha = 0.2$.

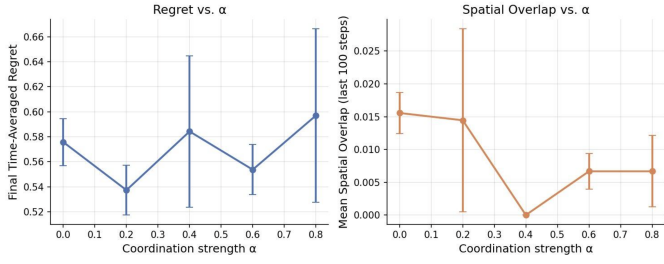


Fig. 5. Preliminary alpha sweep comparing baseline against V1 across $\alpha \in \{0.0, 0.2, 0.4, 0.6, 0.8\}$. Final regret (left) and spatial overlap (right). V1 achieves lower regret at $\alpha \in \{0.2, 0.4\}$, with degradation at $\alpha = 0.8$ due to over-coordination.

D. Belief Update Frequency

Figure 6 shows how belief update period τ (with $\tau_{GP} = \tau_P$) affects regret and last ROI discovery time. Several observations emerge. First, for all conditions, performance degrades significantly at $\tau = 50$ and $\tau = 100$, showing that stale beliefs prevent UAVs from adapting their policies to the current importance map. Second, coordination benefit is most pronounced at $\tau \in \{10, 20\}$: at $\tau = 20$, V2 ($\alpha = 0.2$) achieves lower regret and faster last ROI discovery than the baseline. Third, at high τ , the coordination term loses effectiveness because the neighbor empirical signals $\hat{\rho}_k^j$ are stale and no longer reflect current teammate positions. This suggests that coordination benefit is contingent on maintaining reasonably frequent belief updates. We recommend $\tau \leq 20$ for the coordination mechanism to be effective.

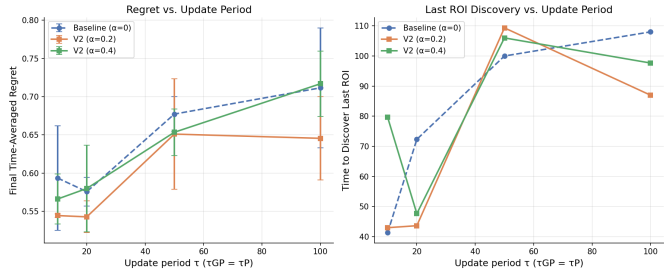


Fig. 6. Effect of belief/policy update period τ on final regret (left) and last ROI discovery time (right) for baseline, V2 $\alpha = 0.2$, and V2 $\alpha = 0.4$. Coordination benefit is robust at $\tau \leq 20$.

E. Scalability

Figure 7 shows results as team size M varies in $\{2, 3, 4, 6\}$. Regret decreases with M for both conditions. V2 consistently achieves lower regret than baseline across all team sizes, with the gap most pronounced at $M = 4$. This suggests coordination benefit grows as the team becomes large enough for spatial overlap to be a meaningful source of inefficiency, but not so large that natural spread already reduces redundancy. Last ROI discovery time drops sharply from $M = 2$ to $M = 4$, with V2 achieving faster discovery than baseline at small team sizes. Spatial overlap does not follow a clear trend with team size and shows high variance across seeds, suggesting the 8×8 grid may be too small relative to team size to reliably measure

this metric at $M \geq 4$. A larger environment would be needed to properly assess spatial overlap at scale.

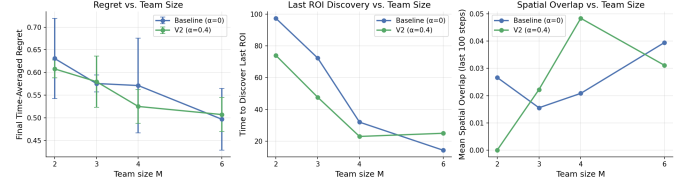


Fig. 7. Scalability across team size $M \in \{2, 3, 4, 6\}$. V2 consistently achieves lower regret and faster last ROI discovery than the baseline.

V. DISCUSSION AND CONCLUSION

Summary. We proposed a coordination-aware modification to the decentralized ergodic coverage framework of Mendoza et al. [2] that personalizes each UAV's target distribution by subtracting a weighted neighbor empirical coverage signal. The importance-weighted formulation V2 reduces spatial overlap by 73%, improves ROI discovery evenness, and scales across team sizes, while maintaining sublinear regret comparable to the baseline.

The V1 artifact and V2's fix. V1 revealed that uniform subtraction is insufficient. It pushes UAVs away from high-importance regions as aggressively as low-importance ones, creating spurious high-visitation areas. V2's importance weighting $(1 - \bar{\rho}_k^m(r))$ directly addresses this. Regions with high belief weight are protected from subtraction, preserving ergodic coverage quality while still diversifying spatial coverage.

Limitations. All experiments use a static information map. Dynamic environments, where ROIs shift or expand over time, are the most practically relevant setting and remain unvalidated. The coordination parameter α was selected empirically; a principled method for adaptive α selection (e.g., based on current GP uncertainty or team spread) would strengthen the approach. The formal regret bound for the coordinated algorithm is left for future work; we conjecture that V2 preserves sublinear regret since normalization guarantees a valid REMC input, maintaining the ergodic property of the induced Markov chain.

Future Work. Priority directions include: (1) evaluating V2 in dynamic environments under Mendoza's two change models (relocation and expansion); (2) studying the effect of varying τ_{GP} and τ_P independently, since the coordination term may be more sensitive to belief update frequency than policy update frequency; (3) developing adaptive α based on current belief uncertainty; and (4) a formal regret analysis for the coordinated algorithm.

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